
Lecture Notes

on Sequences, Recurrence, and Difference Equations

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1 Sequences

1.1 Definition

A **sequence** is a function whose domain is the set of natural numbers ($\mathbb{N}$) and whose range is the set of real numbers ($\mathbb{R}$). Formally, a sequence is written as

$$f: \mathbb{N} \rightarrow \mathbb{R}.$$

$$a_n = f(n), \quad n = 1, 2, 3, \dots$$

where a_n is called the **n -th term** of the sequence.

Notation: A sequence can be written as (a_n) or simply $\{a_n\}$.

$$\{a_n\}_{n=1}^{\infty} \quad \{a_n\}_{n=0}^{\infty}$$

1.2 Examples of Sequences

- $a_n = (-1)^n$: an alternating sequence $1, -1, 1, -1, \dots$

$$n = 0, 1, 2, \dots$$

$$\{(-1)^0, (-1)^1, (-1)^2, \dots\}$$

- $a_n = \frac{1}{n}$: a decreasing sequence $1, \frac{1}{2}, \frac{1}{3}, \dots$

$$n = 1, 2, 3, \dots$$

$$\{1, -1, 1, \dots\}$$

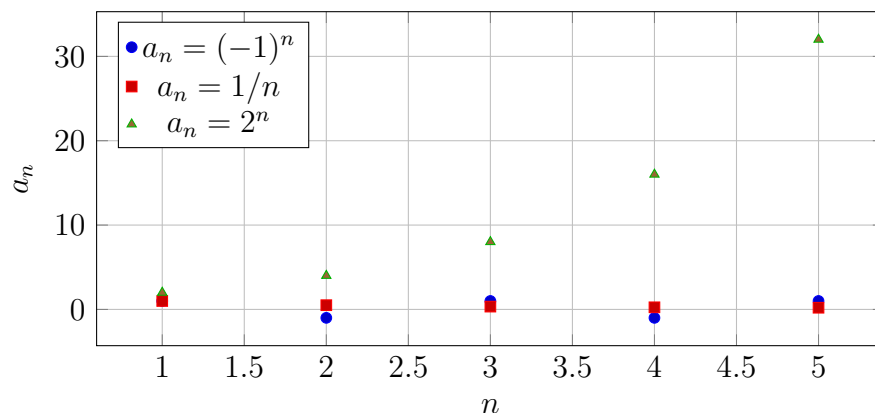
- $a_n = 2^n$: an increasing (exponential) sequence $2, 4, 8, 16, \dots$

$$n = 1, 2, 3, \dots$$

$$\{\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$$

1.3 Graphical Illustration

Examples of Sequences



$$\{2^1, 2^2, 2^3, \dots\}$$

$$[0, 1]$$

☞ **Example:** The n^{th} term of a sequence is given by

$$a_n = f(n) = \frac{(-1)^n 2^n}{(n+1)}.$$

Find the first three terms of the sequence.

$$a_1 = f(1) = \frac{(-1)^1 2^1}{(1+1)} = \frac{-2}{2} = -1.$$

$$a_2 = f(2) = \frac{(-1)^2 2^2}{(2+1)} = \frac{4}{3}$$

$$a_3 = f(3) = \frac{(-1)^3 2^3}{(3+1)} = \frac{-8}{4} = -2.$$

$$\{ 2^2, -2^4, 2^6, -2^8, \dots \}$$

$$\hookrightarrow \text{If } n = 1, 2, \dots \text{ then } a_n = (-1)^{n+1} 2^n$$

$$\text{If } n = 0, 1, 2, \dots \text{ then } a_n = (-1)^n 2^{(n+1)}$$

1.4 Limit of a Sequence

A sequence (a_n) is said to **converge** to $L \in \mathbb{R}$ if

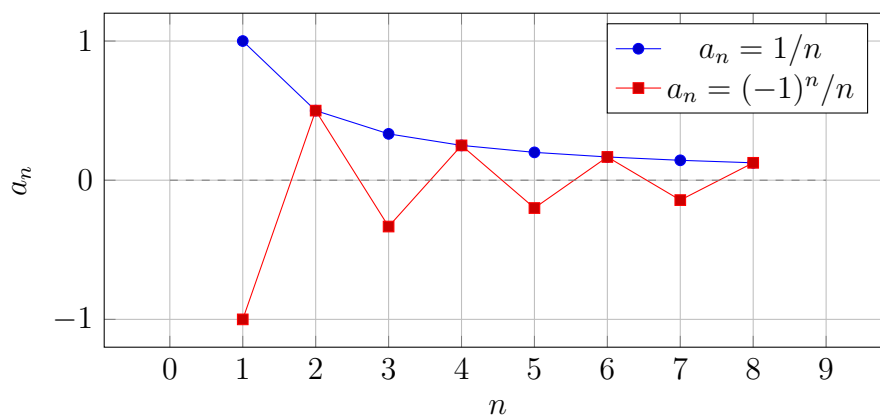
$$\lim_{n \rightarrow \infty} a_n = L.$$

If no such finite limit exists, the sequence **diverges**.

☞ **Examples:**

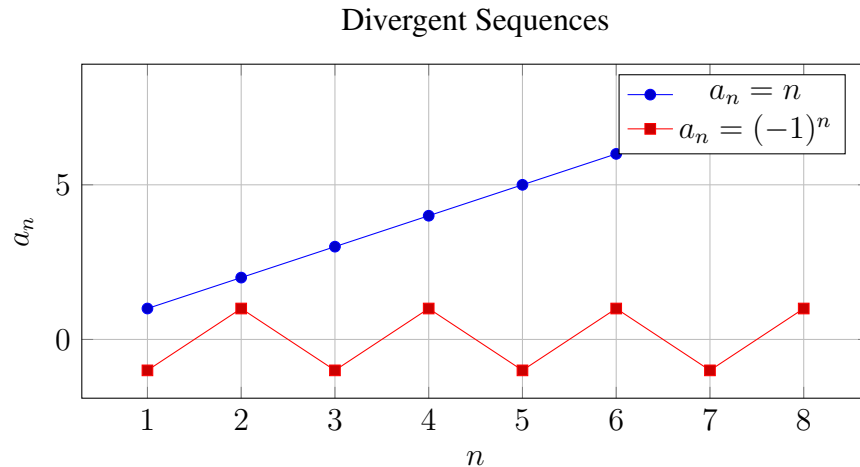
- $a_n = \frac{1}{n}$ converges to 0.
- $a_n = \frac{(-1)^n}{n}$ converges to 0 (oscillatory, but damped).

Convergent Sequences



☛ Examples:

- $a_n = n$ diverges to $+\infty$.
- $a_n = (-1)^n$ oscillates and does not converge.



2 Discrete Difference Equations

2.1 Example 1: Population Doubling

Suppose in an area the population of humans doubles **each year**. Let x_n denote the population in the n^{th} year. Then, by the given information,

$$x_{n+1} = 2x_n \quad (1)$$

which is a **difference equation of order 1**.

Here, the time step is 1 year. If x_0 is known, then we can find the population for any year.

Solution:

From (1), if x_0 is given:

↪ population of humans for current year or Initial population.

↙ 2025-

$$\begin{aligned}
 2026 \quad x_1 &= 2x_0, \\
 2027 \quad x_2 &= 2x_1 = 2(2x_0) = 2^2x_0, \\
 x_3 &= 2x_2 = 2(2^2x_0) = 2^3x_0, \\
 &\vdots \\
 x_n &= 2^n x_0.
 \end{aligned}$$

2026 $x_1 = 2x_0$
 2027 $x_2 = 2x_1 = 2(2x_0) = 2^2x_0$
 2028 $x_3 = 2x_2 = 2(2^2x_0) = 2^3x_0$
 $\vdots$

Thus, the solution for the difference equation (1) is

$x_n = 2^n x_0.$

Equation (1) is sometimes also called a **recurrence relation**. This type of equation is known as a **discrete difference equation**.

2.2 Example 2: Population as the Sum of the Previous Two Years

Now suppose the population in year $n + 1$ is the sum of the populations in the two previous years. Then the difference equation is

$$x_{n+1} = x_n + x_{n-1}.$$

$$\underline{x_0, x_1} \quad (2)$$

This is a **difference equation of order 2**, since two previous terms are required.

$$x_2 = x_1 + x_0$$

$$x_3 = x_2 + x_1$$

Remark:

If only x_0 is given, we cannot solve for x_{n+1} uniquely. We also need x_1 .

2.3 General Form

In general, an m^{th} order difference equation is written as

$$x_{n+1} = f(x_n, x_{n-1}, x_{n-2}, \dots, x_{n-(m-1)}),$$

or equivalently,

$$x_n = f(x_{n-1}, x_{n-2}, \dots, x_1, x_0).$$

 **Example:** A population of doves increases by 3% each year. Let x_n be the size of the population in year n .

1. What is the difference equation?
2. What is the order of the equation? **1**
3. Find the general solution to the difference equation.
4. Find the value of the dove population at the end of 2 years where the initial population is given as 100.

$$\begin{array}{r} 10 \\ \times 100 \\ \hline 1000 \\ 100 \\ \hline 1100 \\ = 1100 \end{array}$$

1) Given x_n denotes the size of the population in year n .
So, the next year the population will increase by

$\frac{3}{100} \times x_n$
 $\therefore$ On the $(n+1)^{\text{th}}$ year,

$$x_{n+1} = x_n + \frac{3}{100} x_n$$

$$= \left(1 + \frac{3}{100}\right) x_n$$

$$= \left(\frac{103}{100}\right) x_n = 1.03 x_n. \quad (*)$$

Equⁿ (*) is the required difference equⁿ.

2) The order of the difference eqn (*) is 1.

$$\begin{aligned} 3) \quad & \text{For } n=0, & x_1 &= 1.03x_0 \\ & n=1, & x_2 &= 1.03x_1 = 1.03(1.03x_0) = (1.03)^2x_0 \\ & n=2, & x_3 &= 1.03x_2 = (1.03)(1.03)^2x_0 = (1.03)^3x_0 \\ & & \vdots & \\ & & x_n &= (1.03)^nx_0. \end{aligned}$$

Thus the general solution of the difference eqn (*) is

$$x_n = (1.03)^nx_0.$$

4) Given, $x_0 = 100$, need to find $x_2 = ?$

$$\therefore x_2 = (1.03)^2 100 \approx 1.06 \times 100 = 106$$